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compact1, compact2, compact3

java.lang

Class StackTraceElement

java.lang.Object
    java.lang.StackTraceElement

All Implemented Interfaces:

Serializable

```
public final class StackTraceElement
extends Object
implements Serializable
```

An element in a stack trace, as returned by `Throwable.getStackTrace()`. Each element represents a single stack frame. All stack frames except for the one at the top of the stack represent a method invocation. The frame at the top of the stack represents the execution point at which the stack trace was generated. Typically, this is the point at which the throwable corresponding to the stack trace was created.

Since:

1.4

See Also:

Serialized Form

Constructor Summary

Constructors

Constructor
`StackTraceElement(String declaringClass, String methodName, String fileName, int`

Method Summary

Concrete Methods

Modifier and Type	Method	Description
boolean	**equals**(**Object** obj)	Returns true if the specified object is another `StackTraceElement` instance representing the same execution point as this instance.
String	**getClassName**()	Returns the fully qualified name of the class containing the execution point represented by this stack trace element.
String	**getFileName**()	Returns the name of the source file containing the execution point represented by this stack trace element.
int	**getLineNumber**()	Returns the line number of the source line containing the execution point represented by this stack trace element.
String	**getMethodName**()	Returns the name of the method containing the execution point represented by this stack trace element.
int	**hashCode**()	Returns a hash code value for this stack trace element.
boolean	**isNativeMethod**()	Returns true if the method containing the execution point represented by this stack trace element is a native method.
String	**toString**()	Returns a string representation of this stack trace element.

Methods inherited from class java.lang.Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Constructor Detail

StackTraceElement

```
public StackTraceElement(String declaringClass,
                         String methodName,
                         String fileName,
                         int lineNumber)
```

Creates a stack trace element representing the specified execution point.

Parameters:

`declaringClass` - the fully qualified name of the class containing the execution point represented by the stack trace element

`methodName` - the name of the method containing the execution point represented by the stack trace element

`fileName` - the name of the file containing the execution point represented by the stack trace element, or null if this information is unavailable

`lineNumber` - the line number of the source line containing the execution point represented by this stack trace element, or a negative number if this information is unavailable. A value of -2 indicates that the method containing the execution point is a native method

Throws:

`NullPointerException` - if declaringClass or methodName is null

Since:

1.5

Method Detail

getFileName

```
public String getFileName()
```

Returns the name of the source file containing the execution point represented by this stack trace element. Generally, this corresponds to the

SourceFile attribute of the relevant class file (as per *The Java Virtual Machine Specification*, Section 4.7.7). In some systems, the name may refer to some source code unit other than a file, such as an entry in source repository.

Returns:

the name of the file containing the execution point represented by this stack trace element, or null if this information is unavailable.

getLineNumber

```
public int getLineNumber()
```

Returns the line number of the source line containing the execution point represented by this stack trace element. Generally, this is derived from the LineNumberTable attribute of the relevant class file (as per *The Java Virtual Machine Specification*, Section 4.7.8).

Returns:

the line number of the source line containing the execution point represented by this stack trace element, or a negative number if this information is unavailable.

getClassName

```
public String getClassName()
```

Returns the fully qualified name of the class containing the execution point represented by this stack trace element.

Returns:

the fully qualified name of the Class containing the execution point represented by this stack trace element.

getMethodName

```
public String getMethodName()
```

Returns the name of the method containing the execution point represented by this stack trace element. If the execution point is contained in an instance or class initializer, this method will return the appropriate *special method name*, <init> or <clinit>, as per Section 3.9 of *The Java Virtual Machine Specification*.

Returns:

the name of the method containing the execution point represented by this stack trace element.

isNativeMethod

```
public boolean isNativeMethod()
```

Returns true if the method containing the execution point represented by this stack trace element is a native method.

Returns:

true if the method containing the execution point represented by this stack trace element is a native method.

toString

```
public String toString()
```

Returns a string representation of this stack trace element. The format of this string depends on the implementation, but the following examples may be regarded as typical:

- `"MyClass.mash(MyClass.java:9)"` - Here, `"MyClass"` is the *fully-qualified name* of the class containing the execution point represented by this stack trace element, `"mash"` is the name of the method containing the execution point, `"MyClass.java"` is the source file containing the execution point, and `"9"` is the line number of the source line containing the execution point.
- `"MyClass.mash(MyClass.java)"` - As above, but the line number is unavailable.
- `"MyClass.mash(Unknown Source)"` - As above, but neither the file name nor the line number are available.
- `"MyClass.mash(Native Method)"` - As above, but neither the file name nor the line number are available, and the method containing the execution point is known to be a native method.

Overrides:

`toString` in class `Object`

Returns:

a string representation of the object.

See Also:

`Throwable.printStackTrace()`

equals

`public boolean equals(Object obj)`

Returns true if the specified object is another `StackTraceElement` instance representing the same execution point as this instance. Two stack trace elements `a` and `b` are equal if and only if:

```
equals(a.getFileName(), b.getFileName()) &&
a.getLineNumber() == b.getLineNumber()) &&
equals(a.getClassName(), b.getClassName()) &&
equals(a.getMethodName(), b.getMethodName())
```

where `equals` has the semantics of `Objects.equals`.

Overrides:

`equals` in class `Object`

Parameters:

`obj - the object to be compared with this stack trace element.`

Returns:

`true if the specified object is another StackTraceElement instance representing the same execution point as this instance.`

See Also:

`Object.hashCode()`, `HashMap`

hashCode

`public int hashCode()`

Returns a hash code value for this stack trace element.

Overrides:

`hashCode` in class `Object`

Returns:

`a hash code value for this object.`

See Also:

```
Object.equals(java.lang.Object),
System.identityHashCode(java.lang.Object)
```


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